



Interactions in Informational Text

Grade 7 • North Carolina • Reading

RI.7.3

Analyze how individuals, events, and ideas interact to produce the development of the polio vaccine. An 80-minute block-period packet with warm-up, guided practice, partner discussion, independent practice, small-group work, and a full answer key.

PRINT • TEACH • REVISIT



Plan the 80-minute block

Level and alignment

Grade 7 · North Carolina · Reading; RI.7.3 — Analyze the interactions between individuals, events, and ideas in a text (e.g., how ideas influence individuals or events, or how individuals influence ideas or events).

What is included

Cover; teacher guide; vocabulary and lesson preview; original passage; warm-up (2 items); guided practice with organizer (4 items); partner discussion (3 items); independent practice (6 items); small-group intervention and enrichment (4 items); exit ticket; complete answer key; access and sources page.

Suggested pacing (80 minutes)

Warm-up 8 min · passage + guided practice 25 min · partner discussion 12 min · independent practice 20 min · small-group work 10 min · exit ticket 5 min.

Preparation

Print student pages separately from the answer key. Review the passage and organizer before class. No outside novel or paid app required.

Student directions

Read the passage carefully. Use evidence from the text to support every answer. Quote or closely paraphrase the text and name the paragraph.

Before you read

Key vocabulary

Lesson focus

Informational texts about history and science often show individuals, events, and ideas affecting each other, not existing separately. An idea can push a person to act; a person's action can create an event; an event can change how people think about an idea.

Model 1 — An outbreak (event) causes a scientist (individual) to change her research focus (idea). What kind of interaction is this?

An event influencing an individual, which then changes the direction of an idea (her research focus).

Model 2 — A new theory (idea) convinces a government to fund a research program (event). What kind of interaction is this?

An idea influencing an event, since the theory directly led to a concrete action (funding the program).



WARM-UP

Warm-up · Before you read

Answer using what you already know. Five minutes.

1. What individual is introduced in paragraph 2, and what idea is he working on?

2. What event does paragraph 1 describe, and what were its effects?



Racing an Invisible Enemy

1. In the summer of 1952, the United States recorded its worst polio outbreak in history: nearly 58,000 cases, many resulting in paralysis, some in death. Public pools closed. Parents kept children indoors. The virus, which attacked the nervous system, could not be reliably treated, only prevented — and no one had yet developed a way to prevent it.
2. Jonas Salk, a virologist at the University of Pittsburgh, had spent the previous several years developing a vaccine using an unconventional approach: a "killed-virus" method, using inactivated poliovirus that could not cause infection but might still train the immune system to recognize it.
3. Many of Salk's colleagues favored a competing idea: a "live-virus" vaccine, using a weakened but still-living form of the virus, which some researchers believed would produce stronger, longer-lasting immunity. The scientific community was divided, and the 1952 outbreak added urgency to a debate that had, until then, moved slowly.
4. The scale of the outbreak influenced funding decisions directly. The National Foundation for Infantile Paralysis, already collecting small donations from millions of Americans through its March of Dimes campaign, redirected significant resources toward accelerating vaccine trials.
5. In 1954, backed by that funding, Salk's killed-virus vaccine entered a massive field trial — nearly two million children participated, making it one of the largest medical experiments in history to that point. The scale of the outbreak had made the scale of the trial possible; without the public fear and urgency 1952 created, funding on that level would have been far harder to secure.
6. The trial's results, announced in April 1955, showed the vaccine was safe and effective. The announcement triggered a new event: a national rollout so urgent that some manufacturers rushed production, leading to a separate, serious problem when one improperly prepared batch caused a small number of actual polio cases before stricter safety protocols were enforced.
7. Cases in the United States fell from tens of thousands per year in the early 1950s to under a thousand by the early 1960s. The idea that had once divided the scientific community — a killed, inactivated virus could safely produce immunity — was, by then, no longer in question.



Guided practice · Evidence organizer

Work through this organizer together before independent practice.

It Says

Quote or closely paraphrase the text.

It Means

Explain the evidence in your own words.

And So

State the inference or analysis it supports.

1. How did the event in paragraph 1 interact with the idea described in paragraph 3?

2. What competing idea does paragraph 3 introduce, and who favored it?

3. How did the event (the outbreak) influence funding, according to paragraph 4?

4. How did that funding (influenced by the event) affect Salk's work, according to paragraph 5?

Partner discussion

Discuss each question with a partner before writing a joint response.

1. Explain the chain of interaction from the 1952 outbreak to the 1954 trial, using evidence from at least two paragraphs.

2. What event does paragraph 6 describe as resulting from the trial's success?

3. How did the event in paragraph 6 (the rushed rollout) create a new problem?



Independent practice

Work independently. Cite evidence from the text in every answer.

1. How did that problem likely influence future individuals or events, based on the text's wording?

2. How did Salk's idea (paragraph 2) ultimately influence public health outcomes, according to paragraph 7?

3. How does paragraph 7 show an idea changing because of events?



Independent practice

Work independently. Cite evidence from the text in every answer.

4. Identify one place in the text where an individual's idea directly caused an event.

5. Identify one place in the text where an event directly caused a change in an idea's acceptance.

6. How might the outcome have been different if the 1952 outbreak had not occurred, based on the text's cause-and-effect pattern?



Small-group work

Use with students who need additional support.

1. Explain how the March of Dimes campaign (paragraph 4) connects an event, an individual effort, and an idea.

2. Why does the text mention the competing live-virus idea in paragraph 3 instead of only describing Salk's approach?

Small-group work

Use with students ready for additional challenge.

1. How do individuals, events, and ideas each play a distinct role in this passage? Give one example of each.

2. Trace one idea across the entire passage, from its introduction to its final outcome.



Exit ticket

Answer independently to show what you learned today.

1. Explain, using two pieces of evidence, how an event and an idea interacted to produce a specific outcome in this text.



Exit ticket · self-assessment

Rate today's work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this with my teacher.



Answer key

1. Jonas Salk, a virologist developing a "killed-virus" vaccine using inactivated poliovirus.
2. The 1952 polio outbreak, which caused nearly 58,000 cases and led to closed pools and parents keeping children indoors.
3. The outbreak added urgency to the scientific debate between the killed-virus and live-virus approaches, a debate paragraph 3 says had "moved slowly" before 1952.
4. A "live-virus" vaccine using a weakened but living form of the virus, favored by many of Salk's colleagues.
5. The National Foundation for Infantile Paralysis redirected significant resources from its March of Dimes campaign toward accelerating vaccine trials.

Accept equivalent, well-supported alternative answers throughout.



Answer key

6. It allowed his killed-virus vaccine to enter a massive 1954 field trial involving nearly two million children.

7. The 1952 outbreak (event) created urgency and fear, which led the Foundation to redirect funding (paragraph 4), and that funding then made the large-scale 1954 trial possible (paragraph 5) — an event influencing funding, which then enabled a further event.

8. A national rollout of the vaccine, announced after the trial's results were released in April 1955.

9. Some manufacturers rushed production, and one improperly prepared batch caused a small number of actual polio cases before stricter safety protocols were enforced.

10. It likely influenced regulators or manufacturers to enforce "stricter safety protocols," suggesting the problem directly caused a change in vaccine production oversight.

Accept equivalent, well-supported alternative answers throughout.



Answer key

11. Cases fell from tens of thousands per year in the early 1950s to under a thousand by the early 1960s, showing the vaccine's idea translated into a major real-world effect.

12. It states the once-divided scientific community no longer questioned "the idea that... a killed, inactivated virus could safely produce immunity," showing the idea gained acceptance because of the trial's results and the vaccine's real-world success.

13. Salk's idea for a killed-virus vaccine (paragraph 2) directly led to the 1954 field trial (paragraph 5), an event testing that idea.

14. The 1955 trial results (event) caused the killed-virus idea to become widely accepted, ending the earlier scientific debate described in paragraph 3.

15. Funding might not have been redirected as urgently (paragraph 4), which could have delayed or shrunk the 1954 trial, since the text ties the trial's scale directly to the outbreak's urgency.

Accept equivalent, well-supported alternative answers throughout.



Answer key

16. The campaign was an ongoing public effort (individuals donating) that became redirected because of the outbreak (event) to fund Salk's vaccine idea, showing all three elements working together.

17. It shows that Salk's idea was not the only option being considered, which helps explain why the urgency from the 1952 outbreak (forcing a faster decision) mattered so much to the scientific debate.

18. Individuals: Jonas Salk developing the vaccine. Events: the 1952 outbreak and the 1954 trial. Ideas: the killed-virus versus live-virus vaccine approaches.

19. The killed-virus idea is introduced in paragraph 2, debated in paragraph 3, tested at scale in paragraph 5, proven effective in paragraph 6, and fully accepted by paragraph 7 — moving from a contested idea to an established fact.

20. Answers vary; require two specific, correctly cited details showing a clear cause-and-effect interaction between an event and an idea (or individual).

Accept equivalent, well-supported alternative answers throughout.



Access, sources, and license

AIG

Ask for a second interpretation, a counterexample, or a student-written item at the same rigor, with an explanation of the choice.

504

Follow the individual plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

EC

Pre-teach key vocabulary before reading. Model the first guided-practice item aloud. Accept oral or dictated responses when appropriate.

ESL

Preview key vocabulary with examples. Allow brief home-language discussion before independent writing. Provide sentence starters for open-response items.

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RI.7.3 — Analyze the interactions between individuals, events, and ideas in a text (e.g., how ideas influence individuals or events, or how individuals influence ideas or events).. The objective is original wording describing a focused part of the standard.

NC ELA Standard Course of Study, Grade 7: <https://www.dpi.nc.gov/documents/publications/catalog/>

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